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Modeling the role and impact of alien species and fisheries on the Israeli marine continental shelf ecosystem

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ABSTRACT

The ecosystems of the Israeli Mediterranean coast have undergone significant changes in recent decades mainly due to species invasions and fishing. In order to characterize the structure and functioning of the marine continental shelf of the Israeli Mediterranean coast and assess temporal changes, we developed a food web model representing two time periods: 1990–1994 and 2008–2012.

The 1990–1994 and 2008–2012 food web models were composed of 39 and 41 functional groups, respectively. Functional groups ranged from primary producers to top predators, and included six and eight alien functional groups, respectively, encompassing several crustacean and fish species. Input data included local surveys and fishery statistics, published data on stomach content analyses, and the application of empirical equations to estimate consumption and production rates.

Results of the competitive interactions between alien and native species and changes in trophic flows between food web components highlight the increasing impact of alien species over time. Fishing had noticeable impacts in both time periods and played an important role in the ecosystem. Despite different productivity rates and other environmental differences, the Israeli marine ecosystem shared common structural and functional traits with other Mediterranean marine ecosystems. This is the first attempt to study the ecosystem of the Levant region using mass-balance models and to integrate such a large amount of alien species into food web analyses.

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1. Introduction

Biological invasions are currently considered as one of the most important direct drivers of biodiversity change and pose a major threat on marine ecosystems, with both ecological and economic impacts (Bax et al., 2003; Molnar et al., 2008; Rilov and Crooks, 2009). These impacts include changes in the genetic diversity of native species, local species extinctions and alterations of entire food webs (Grosholz, 2002; Levin and Crooks, 2011).

The Mediterranean Sea, specifically its eastern basin (Rilov and Galil, 2009; Edelist et al., 2013a; Katsanevakis et al., 2014), is one of the marine ecosystem most affected by alien species (Molnar et al., 2008; Costello et al., 2010). Currently, there are nearly 700 alien species described in the Mediterranean Sea, of which more than half are considered to be established and spreading (Zenetos et al., 2010; Galil et al., 2014; Galil et al., 2016). Although no complete extinctions, due to alien species, have been recorded in the Mediterranean Sea, sudden declines in abundance and local extirpations of native species concurrent with the proliferation of alien species have been documented (Galil, 2007a; Golani, 2010; Edelist et al., 2013a).

The Mediterranean Sea is currently classified among the most impacted ecoregions of the world (Costello et al., 2010; Halpern et al., 2015). The increasing impact of fishing in the Mediterranean Sea is evident. Many demersal and pelagic stocks are fully exploited or overexploited (Vasilakopoulos et al., 2014; Tsikliras et al., 2015). In addition, the Mediterranean Sea is being altered by other anthropogenic activities posing impacts, such as habitat loss and degradation, pollution, eutrophication and climate change (Coll et al., 2010; Coll et al., 2012).

Within this context, the marine ecosystem of the Israeli Mediterranean coast, located within the Levantine Sea in the eastern part of the Mediterranean basin, has undergone significant changes in recent decades caused primarily by species invasion, fishing, river damming and climate change (Edelist et al., 2013a; Goren et al., 2013; Edelist et al., 2014; Sternberg et al., 2015).

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The Levantine Sea has the hottest, most saline and most oligotrophic waters in the Mediterranean Sea (Azov, 1991; Brasseur et al., 1996), as a result of high evaporation rates, very low riverine inputs and limited vertical mixing. These characteristics are the main factors influencing several faunistic phenomena such as "Levantine nanism" (Sonin et al., 2007) and general faunal impoverishment compared to other Mediterranean Sea regions (Coll et al., 2010).

Multiple empty niches in the Levant due to the low regional biodiversity and the existence of the native species in a habitat at the limits of their tolerance levels (Golani, 1998; Galil, 2008; Rilov and Galil, 2009) can be potentially utilized by alien species. In addition, the opening of the Suez Canal in 1869, its continuous enlargement and the similar temperature and salinity regimes in the Levantine Sea and the Red Sea allowed for the progressive introduction of many Indo-Pacific species into the eastern Mediterranean Sea (known as Lessepsian immigrants) (Rilov and Galil, 2009). This phenomenon is almost entirely unidirectional, i.e. into the Mediterranean, and it was accelerated during the second half of the 20th century and the first decade of the 21th century (Rilov and Galil, 2009; Golani, 2010).

The environment of the eastern Mediterranean Sea has become more extreme to its native species due to climate change, favouring the establishment and spread of thermophilic species, such as most of the Lessepsian immigrants (Ben Rais Lasram et al., 2010; Lejeusne et al., 2010; Bianchi et al., 2013). Currently, the Levantine Sea is the world's most invaded marine ecoregion and the ecosystem has shifted considerably due to the collapse of several native species and the increasing dominance of alien species (Edelist et al., 2013a).

The study of biological invasions and their impacts on the marine environment has increased in the last few decades and it is providing new insights into their ecological consequences (Wonham and Lewis, 2009). Most studies of alien species on marine ecosystems have focused on population level questions, such as population establishment, spread, impacts, invasion control and spatial distributions (e.g., Ruesink and Collado-Vides, 2006; Azzurro et al., 2013). In addition, several tools of different complexity have been employed to examine the impact of alien species (Wonham and Lewis, 2009). These applications range from analyses that include a number of groups within the pelagic compartment to modeling applications that include entire food webs (e.g., Berdnikov et al., 1999; Oguz et al., 2001; Arias-González et al., 2011; Libralato et al., 2015).

Given the interconnectedness between populations, environment and human activities, a shift towards a more comprehensive analysis and management of human activities is required, as it is emphasized by the ecosystem-based management (EBM) approach (Rosenberg and McLeod, 2005; Link, 2010a). Within this context, ecosystem modeling tools are particularly useful because they allow the study of marine ecosystems as a whole, integrating available information to study direct and indirect interactions among ecosystem compartments, i.e. trophic interactions and the impact of fishing activity on marine resources (Plagányi, 2007; Fulton, 2010). One popular modeling approach used to study marine food webs is the Ecopath with Ecosim (EwE) approach (Christensen and Walters, 2004), widely applied to describe the structure and functioning of aquatic food webs and to assess the impacts of human activities and environmental changes (Christensen and Walters, 2011; Heymans et al., 2014; Colléter et al., 2015; Heymans et al., 2016).

Several Ecopath models have been built over the last decade in the Mediterranean Sea (Coll and Libralato, 2012). However, all available ecosystem models (with the exception of the Gulf of Gabes (Hattab et al., 2013), in Tunis) were developed for the northern Mediterranean Sea, and mostly in the western basin. In addition, the EwE approach has been used to study the impacts of alien species in several aquatic systems but typically, these studies have included only one or two alien functional groups (Corrales et al., 2014).

In this study we developed a food web Ecopath model of the Israeli Mediterranean continental shelf for two time periods (1990–1994 and 2008–2010). The models were constructed in order to characterize the structure and functioning of the food web and to assess the past and current impact of several alien species and fishing on the Israeli continental shelf food web. Due to the large number of alien species in the region, this represents a major modeling challenge and an important step forward in modeling alien species with EwE, generally, and the Mediterranean Sea in particular.

2. Material and methods

2.1. Study area and time periods

The Israeli Mediterranean continental shelf ecosystem (thereafter referred to ICS) (Fig. 1) food web models cover an area of 3725 km^2 between 0 and 200 m depth. The bathymetric extension of the models is constrained by biology and fisheries as the study area includes the entire continental shelf, where the alien species live, and where the fishery mainly operates. Two time periods were modeled: 1990–1994 (thereafter referred to 1990s) and 2008–2012 (thereafter referred to 2010s). These two time periods were chosen due to the availability of data, the increase in the extent of occurrence of alien species, and the decline in fishing from 1990 to present (Edelist et al., 2013a; Goren et al., 2013).

Israel's Mediterranean fisheries include a small semi-industrial trawl fleet, a small artisanal sector of gill and trammel-netters and longliners, a small purse seine fleet and a growing recreational sector (Edelist et al., 2013b; Goren et al., 2013). They are all multi-species fleets (Edelist et al., 2013b), that fish on the continental shelf, mainly near the coast (Edelist et al., 2011).

2.2. Ecopath modeling approach

Ecopath with Ecosim (EwE) version 6.4.3 (Christensen and Walters, 2004; Christensen et al., 2008) was used to describe ICS food web. We used the static Ecopath model, to provide a quantitative representation of the food web as a "snapshot" in terms of flows and biomasses for a defined period of time. The food web is modeled by using functional groups, which can consist of ontogenetic fractions of a species, single species or groups of species sharing common ecological traits. These groups are linked through their trophic interactions.

Ecopath is based on two master equations describing the production (Eq. (1)) and consumption (Eq. (2)) of each functional group (Christensen and Walters, 2004; Christensen et al., 2008).

$$\begin{aligned}\text{Production} = {} & \text{predation mortality} + \text{fishing mortality} \\ & + \text{other mortality} + \text{biomass accumulation} \\ & + \text{net migration}\end{aligned} \quad (1)$$

$$\text{Consumption} = \text{production} + \text{respiration} + \text{unassimilated food} \quad (2)$$

For each functional group, three of the four basic parameters (biomass (B), production (P/B) and consumption (Q/B) rates, and ecotrophic efficiency (EE)) are required and the forth is estimated. In addition, for each functional group the diet composition is required as well as the catch by fleet. A detailed explanation of the algorithms and equations of the approach and its main advantages and limitations are described in Christensen and Walters (2004) and Heymans et al. (2016).

2.3. Functional groups and input data

To represent the ICS food web, we defined the functional groups based on biological and ecological features of species such as habitat, depth distribution, diets, and data availability. We followed a similar food web structure and parametrization to other Ecopath models developed for the Mediterranean Sea (Coll et al., 2006; Tsagarakis et al., 2010; Corrales et al., 2015) (Table 1). The data of the two models, 1990s and 2010s, are shown in Table 1 and a detailed description of these inputs is provided in the Online supplement section 1.

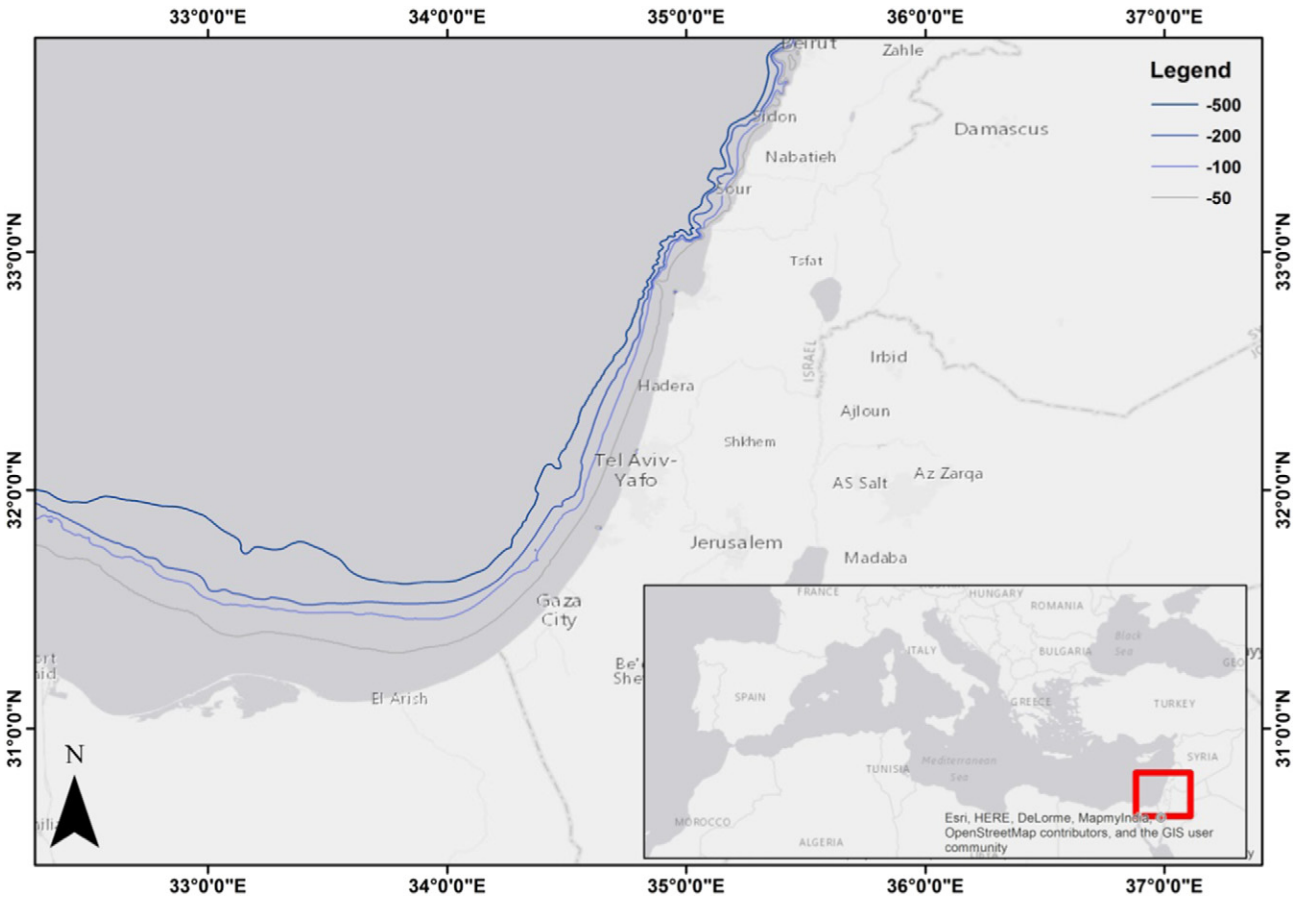


Fig. 1. The study area encompassing the Israeli EEZ and depth contours.

To assess the impact of alien species in the ecosystem we included specific alien functional groups. Due to the large number of alien species and differences in time of settlement in the ecosystem, we included the most abundant species (33 species in 1990s and 41 species in 2010s), for which sufficient data were available, including crustaceans and fish species (see Online supplement section 2). At present, sufficient information for other alien groups such as cephalopods, molluscs, polychaetes and jellyfish is not available.

In addition to ecological traits (feeding habitats and habitat type), we combined alien species into functional groups based on their arrival in the ecosystem. Crustacean species invaded the ecosystem prior to 1990 and were grouped into alien shrimps and alien crabs. Fish species invaded, and continue to invade, the ecosystem both before and after 1990. In the demersal habitat, we defined the following alien fish groups: goatfishes (*Upeneus* sp.), alien herbivores (*Siganus* sp.), alien lizardfish (*Saurida lessepsianus*), earlier alien demersal fishes (demersal fishes that invaded the ecosystem before 1990) and new alien demersal fishes (demersal fishes that invaded the ecosystem after 1990). Small, medium and large pelagic fishes invaded the ecosystem before 1990. However, due to lack of information, these groups had to be combined with native species, creating three groups: small pelagic fishes, native medium pelagic fishes and large pelagic fishes. An additional group of new alien medium pelagic fishes were created for new species that arrived post 1990.

In total, 39 functional groups were used in the ICS model for the 1990s period, including 2 primary producers, 12 groups of invertebrates, 20 groups of fishes, 1 group of sea turtles, 1 group of seabirds, 1 group of dolphins and 2 groups of detritus (natural detritus or "marine snow" and discards) (Table 1). Two alien fish groups were added to the 2010s model to account for the new arrivals (Table 1).

Input parameters for the species and functional groups were mainly obtained from published literature and unpublished information from Israel Oceanographic and Limnological Research (IOLR) and Tel-Aviv University. The data were collected from the study area or surrounding areas (see Online supplement section 1 for details on parameterization of each functional group).

Biomass estimates were obtained from fishery dependent bottom trawl survey (swept-area method), visual surveys conducted in the area, and additional information from the literature. Biomass estimates for pelagic species were not available and therefore we used realistic EE values to estimate the biomass of 10 functional groups (Heymans et al., 2016). Production and consumption rates were either estimated using empirical equations (Heymans et al., 2016) accounting for changes in water temperature between both time periods, or taken from literature or from other models and corrected for temperature changes following Opitz (1996). Diet information was compiled using published and unpublished data on stomach content analyses, giving preference to local or similar areas. To date, limited information on the diet of alien species is available. In the absence of information from the Mediterranean Sea, we assumed the same diet composition for alien crustaceans (shrimps and crabs) as for the native groups. For migratory species (large pelagic fishes, sea birds, turtles and dolphins), we set a fraction of the diet composition as import based on the time that these species feed outside the system (Christensen et al., 2008; Heymans et al., 2016).

Fisheries data were obtained from a reconstruction of Israeli catches including both commercial and discards, the recreational fishery and unregulated and unreported catch (Edelist et al., 2013b). We considered three commercial fishing fleets: bottom trawl, purse seine and artisanal fisheries (including gillnets and longliners).

Table 1
Initial (in bold) and modified input data of the Israeli Mediterranean continental shelf model for 1990–1994 (1990s) and 2008–2012 (2010s) time periods. B = final biomass ($t \cdot km^{-2}$); P/B = production/biomass ($year^{-1}$); Q/B = consumption/biomass ($year^{-1}$); EE = ecotrophic efficiency; P/Q = production/consumption ratio; landings and discards ($t \cdot km^{-2} \cdot year^{-1}$). (*) indicates groups that were considered within the pelagic compartment.

	Functional group	B		P/B		Q/B		EE		P/Q		Landings		Discards	
		1990s	2010s	1990s	2010s	1990s	2010s	1990s	2010s	1990s	2010s	1990s	2010s	1990s	2010s
1	Phytoplankton*	**2.78**	**3.04**	**88.42**	**78.90**	–	–	0.24	0.23			–	–	–	–
2	Benthic primary producers	0.08	0.07	**5.50**	**5.50**	–	–	**0.50**	**0.70**			–	–	–	–
3	Micro and mesozooplankton*	1.15	1.05	**22.82**	**23.19**	69.15	70.27	**0.95**	**0.95**	**0.33**	**0.33**	–	–	–	–
4	Macrozooplankton*	0.16	0.15	17.15	17.43	**57.17**	**58.09**	**0.95**	**0.95**	**0.30**	**0.30**	–	–	–	–
5	Gelatinous plankton*	0.07	0.06	**15.56**	**15.82**	**56.65**	**57.57**	**0.30**	**0.30**	0.27	0.27	–	–	–	–
6	Polychaetes	0.70	0.72	**5.28**	**5.37**	**22.80**	**23.17**	**0.80**	**0.80**	0.23	0.23	–	–	–	–
7	Suprabenthos	0.21	0.23	**11.84**	**11.88**	**57.12**	**57.31**	**0.80**	**0.80**	0.21	0.21	–	–	–	–
8	Native shrimps	0.12	0.08	**3.10**	**3.11**	**9.76**	**9.80**	0.95	0.95	0.32	0.32	**0.04**	**0.00**	**0.01**	**0.03**
9	Alien shrimps	0.07	0.10	**3.10**	**3.11**	**10.12**	**10.16**	0.95	0.95	0.31	0.31	**0.03**	**0.04**	–	–
10	Native crabs	0.05	0.02	**2.80**	**2.81**	**9.25**	**9.29**	0.95	0.95	0.30	0.30	**0.00**	**0.00**	–	–
11	Alien crabs	0.02	0.09	**2.80**	**2.81**	**9.55**	**9.59**	0.95	0.95	0.29	0.29	–	**0.02**	**0.01**	**0.05**
12	Benthic invertebrates	1.13	1.21	3.27	3.28	**10.90**	**10.94**	**0.80**	**0.80**	**0.30**	**0.30**	–	–	–	
13	Benthic cephalopods	0.04	0.05	**1.90**	**1.91**	**6.80**	**6.82**	0.95	0.95	0.28	0.28	**0.01**	**0.01**	–	–
14	Benthopelagic cephalopods*	0.06	0.07	**2.50**	**2.51**	**26.47**	**26.56**	0.95	0.95	0.09	0.09	**0.01**	**0.02**	–	–
15	Mullets	0.05	0.01	**1.86**	**1.87**	**8.32**	**8.37**	0.96	0.98	0.22	0.22	**0.05**	**0.01**	**0.00**	–
16	Goatfishes	0.03	0.03	**1.88**	**1.89**	**8.54**	**8.60**	0.99	0.99	0.22	0.22	**0.02**	**0.03**	**0.00**	**0.01**
17	Hake	0.05	0.01	**0.89**	**0.89**	**5.80**	**5.84**	0.99	0.97	0.15	0.15	**0.02**	**0.00**	–	
18	Flatfishes	0.02	0.02	**1.52**	**1.53**	**8.23**	**8.26**	0.98	0.96	0.18	0.18	**0.00**	**0.00**	**0.01**	**0.01**
19	Rocky fishes	0.06	0.05	**1.73**	**1.74**	**7.18**	**7.20**	0.95	0.98	0.24	0.24	**0.02**	**0.02**	–	–
20	Small native dem. fishes	0.24	0.21	**1.49**	**1.50**	**6.62**	**6.67**	0.98	0.96	0.23	0.22	**0.12**	**0.13**	**0.03**	**0.04**
21	Large native dem. fishes	0.11	0.02	**1.10**	**1.10**	**6.08**	**6.10**	0.97	0.97	0.18	0.18	**0.08**	**0.01**	**0.00**	**0.00**
22	Alien herbivores	0.01	0.02	**1.72**	**1.73**	6.90	6.92	0.95	0.95	**0.25**	**0.25**	**0.00**	**0.01**	–	–
23	Earlier alien dem. fishes	0.03	0.10	**1.58**	**1.59**	**6.79**	**6.81**	0.99	0.97	0.23	0.23	**0.00**	**0.01**	**0.01**	**0.06**
24	New alien dem. fishes	–	0.11	–	**1.49**	–	**6.67**	–	0.95	–	0.22	–	**0.03**	–	**0.04**
25	Alien lizardfish	0.07	0.03	**1.18**	**1.18**	**7.01**	**7.04**	0.93	0.93	0.17	0.17	**0.05**	**0.03**	**0.00**	–
26	Demersal fishes (upper slope)	0.01	0.01	1.01	1.01	**5.92**	**5.94**	0.96	0.93	**0.17**	**0.17**	–	–	**0.00**	**0.00**
27	Benthopelagic fishes*	0.15	0.13	2.03	2.06	**10.14**	**10.28**	**0.95**	**0.95**	**0.20**	**0.20**	**0.03**	**0.02**	**0.02**	**0.03**
28	Mesopelagic fishes*	0.06	0.05	1.74	1.75	**8.70**	**8.73**	**0.95**	**0.95**	**0.20**	**0.20**	–	–	–	–
29	Demersal sharks	0.02	0.04	**0.70**	**0.70**	4.67	4.68	0.64	0.70	0.15	0.15	**0.01**	**0.02**	**0.00**	**0.00**
30	Rays and skates	0.03	0.05	**0.94**	**0.94**	**5.70**	**5.72**	0.87	0.89	0.16	0.16	**0.01**	**0.03**	**0.01**	**0.00**
31	Small pelagic fishes*	0.70	0.58	2.40	2.45	**11.98**	**12.27**	**0.95**	**0.95**	**0.20**	**0.20**	**0.12**	**0.04**	**0.00**	**0.01**
32	Horse mackerel*	0.13	0.09	1.44	1.46	**8.02**	**8.13**	**0.95**	**0.95**	**0.18**	**0.18**	**0.06**	**0.04**	**0.00**	**0.01**
33	Mackerel*	0.07	0.08	1.42	1.44	**7.88**	**7.99**	**0.95**	**0.95**	**0.18**	**0.18**	**0.02**	**0.02**	–	–
34	Native medium pelagic fishes*	0.10	0.07	0.85	0.86	**7.07**	**7.09**	**0.90**	**0.90**	**0.12**	**0.12**	**0.04**	**0.04**	–	–
35	Alien medium pelagic fishes*	–	0.02	–	0.85	–	**7.11**	–	**0.90**	–	**0.12**	–	**0.01**	–	**0.00**
36	Large pelagic fishes*	**0.06**	**0.06**	**0.90**	**0.90**	**5.35**	**5.38**	0.98	0.99	0.17	0.17	**0.05**	**0.06**	–	–
37	Turtles	0.06	0.06	**0.17**	**0.17**	**2.78**	**2.83**	**0.50**	**0.50**	0.06	0.06	–	–	**0.00**	**0.00**
38	Sea birds	**0.00**	**0.00**	**5.10**	**5.10**	**75.23**	**75.23**	0.01	0.01	0.07	0.07	–	–	**0.00**	**0.00**
39	Dolphins*	**0.01**	**0.01**	**0.09**	**0.09**	**10.95**	**10.95**	0.48	0.48	0.01	0.01	–	–	**0.00**	**0.00**
40	Detritus	**17.81**	**17.95**	–	–	–	–	0.23	0.24	–	–	–	–	–	–
41	Discards	**0.17**	**0.29**	–	–	–	–	0.97	0.49	–	–	–	–	–	–

2.4. Pre-balancing and balancing analyses

An Ecopath model is considered ecologically and thermodynamically balanced under the following conditions: (1) estimated EE < 1 for all functional groups, (2) values of P/Q (production/consumption rate or gross efficiency of food conversion, GE) are between 0.1 and 0.35 with the exception of some fast growing groups, (3) R/A (respiration/food assimilation) < 1, (4) R/B (respiration/biomass) are between 1 and 10 for fishes and higher values for small organisms, (5) NE (net efficiency of food conversion) > GE and (6) P/R (production/respiration) < 1 (Christensen et al., 2008; Heymans et al., 2016) (Table 1 and Online supplement section 3).

Initial results of the ICS models showed that the EE > 1 for 16 (1990s) and 15 (2010s) demersal groups (fish, cephalopods and crustaceans). To achieve mass-balance, we applied a manual mass-balanced procedure following a top-down approach modifying appropriate input parameters (starting from the groups with higher trophic levels) and following the best practice guidelines provided in the literature (Heymans et al., 2016).

In order to ensure that the model parameters obeyed general ecologic principles and to guide the balancing procedure we used the PREBAL analysis (Link, 2010b). This analysis highlighted that some P/B and Q/B values had to be adjusted since they were too low or too high based on their trophic levels. For additional information on the application of the PREBAL procedure see Online supplement section 4.

After testing the model parameters with PREBAL, biomass and diets were readjusted where needed, as in other Ecopath models developed in the Mediterranean Sea (Coll et al., 2006; Tsagarakis et al., 2010; Corrales et al., 2015). We estimated biomass by calculating the average and maximum value for each species and year from the bottom surveys. In the absence of catchability coefficients to correct biomass estimates, maximum values were used to ensure mass-balance. We also adjusted the biomass inputs taking into account the fishing mortality patterns in other Mediterranean areas (demersal fish groups) and using a reasonable value of EE (EE values close but lower than 1) (Heymans et al., 2016), leaving the model to calculate the biomass for some groups (e.g. for crustacean and cephalopods groups). Finally, the diet matrix was slightly adjusted to take into account the abundances of species in the ecosystem (Online supplement section 5).

2.5. Pedigree index and quality of the model

The pedigree routine (Christensen and Walters, 2004) was used to quantify the uncertainty associated with the input parameters and the quality of the models and to validate choices made in balancing the model. For each input datum, we assigned pedigree values to record

the degree of confidence associated with the data. With the information for each functional group, the pedigree index was calculated for the overall model. The pedigree index ranges between 0 (low quality) and 1 (high quality), allowing a description of the quality of the model that can be compared to other models. The confidence intervals for the pedigree analysis and index values used are described in Online supplementary section 6.

To evaluate the quality of the model we also compared some model outputs to the results of independent data. The trophic levels (TL) estimated for the 2010s Ecopath model were compared to the ^{15}N values (‰) estimated by Stable Isotope Analysis (SIA) conducted on the Lebanese coast during 2011–2012 (Fanelli et al., 2015). For those functional groups from the model with several ^{15}N values, the values were weighted using biomass proportions of these species in each functional group. TLs estimated by the Ecopath model were plotted against the ^{15}N values and the correlation was tested using the Spearman-rank non-parametric correlation coefficient test (Zar, 1984).

2.6. Model analyses and ecological indicators

In order to assess the changes between the two models and the possible role and impact of alien species and fisheries a number of key food web indicators were analyzed and compared between time periods. In addition, we compared our results with those of the south Catalan Sea (Coll et al., 2006), the northern and central Adriatic Sea (Coll et al., 2007) and the north Aegean Sea (Tsagarakis et al., 2010).

Although the topology (number of functional groups, distribution of the species, area modeled and depth range included) of the model affects many food web indicators (Heymans et al., 2014; Heymans et al., 2016), we assumed that the comparison was possible as both ICS models (1990s and 2010s) represent the same ecosystem with the same model structure with the exception of two new functional groups that invaded the ecosystem during the time (comparison between ICS models). Comparison with other Mediterranean models included only robust indicators following previous studies (Heymans et al., 2014; Corrales et al., 2015). The ICS models were developed following similar criteria to other Mediterranean models (number of functional groups, distribution of the species and parameterization) and included similar depth ranges.

2.6.1. Ecosystem structural and functioning traits

To represent the biomasses, trophic flows and trophic levels (TL), we used a flow diagram and a Lindeman spine representation (Lindeman, 1942). In the Lindeman spine, the flows and biomasses are aggregated for each discrete trophic level and the detritus box is separated from the primary producers to show the amount of energy that flows through it.

The Ecopath model outputs a variety of indicators related to the development and maturity of ecosystems according to Odum (1969). In this study, the following indicators were included: (1) Total system throughput (TST, $t \cdot km^{-2} \cdot year^{-1}$), estimated as the total flows in the ecosystem (sum of all consumption exports, respiration and flow to detritus; (2) total biomass (TB, $t \cdot km^{-2}$); (3) total primary production/total respiration (Pp/R); (4) total primary production/total biomass (Pp/B, $t \cdot km^{-2} \cdot year^{-1}$); (5) system omnivory index (SOI), (6) Finn's cycling index (FCI, %) and (7) Finn's mean path length (PL) (Odum, 1969; Finn, 1976; Christensen, 1995; Christensen et al., 2008; Heymans et al., 2014). We also included the transfer efficiency (TE), the fraction of total flows of each discrete trophic level that are either exported out of the ecosystem (e.g., by the fishing activity) or transferred to higher trophic levels through consumption. TE summarizes the inefficiency or energy dissipation between discrete trophic levels that is produced along the food web due to respiration, excretion, egestion, natural mortality and exports. The mean trophic level of the community (mTLco), calculated by weighting the TL of each functional group by its biomass was also used to compare the models.

2.6.2. Ecological roles of functional groups

Trophic level (TL) was used to analyze the ecological position of the functional groups of the ICS model (Lindeman, 1942; Stergiou and Karpouzi, 2001).

The Mixed Trophic Impact (MTI) analysis was used to quantify the direct and indirect impact in the food web that a hypothetical increase in the biomass of one functional group would have on the biomasses of all the other functional groups in the food web, including the fishing fleets (Ulanowicz and Puccia, 1990; Christensen et al., 2008).

The keystoneness index developed by Valls et al. (2015) was used to identify the keystone species. A keystone species is defined as a predator species with a high and wide impact on the food web despite its low biomass (Paine, 1966, 1969; Valls et al., 2015). The index is calculated as:

$$KS = IC \times BC \quad (3)$$

where IC is the impact component and BC is the biomass component. The IC represents the overall effect of group (i) on all other groups in the food web excluding the impact on (i) itself and the impacts on dead groups and fleets (Libralato et al., 2006). The BC is defined as the rank of the group according to their biomass values in descending order (Valls et al., 2015).

2.6.3. Impacts and role of alien species

To evaluate the impact of alien species on the ecosystem, the changes in the contribution of alien species to the biomass and catch, and changes to flows related to alien species between the two time periods were calculated. For the changes to flows related to alien species, the consumption by alien groups on their prey and the predation on alien groups by their predators for both time periods were estimated.

2.6.4. Impacts of fishing on the food web

The impacts of fishing on the food web were assessed using the mean trophic level of the catch (mTLc) (Christensen, 1996) and the primary production required to sustain the fisheries (% PPR) (Pauly and Christensen, 1995), taking into account both primary and detritus production. The MTI analysis was also used to quantify the direct and indirect impact of each fleet on the functional groups and the exploitation rate (fishing mortality/total mortality, F/Z) was calculated for the functional groups that were exploited.

3. Results

3.1. Ecosystem structural and functioning traits

The overall trophic flows among functional groups and TLs show that in the 1990s most of the flows to detritus (93.3%), consumption (69%) and production (95.8%) were related to the pelagic compartment (Tables 1 and 2) due to the contribution of planktonic groups. Similar contributions (92.8%, 66.1% and 95.4%, respectively) were found in the 2010 model. The flow diagram indicates an important bento-pelagic coupling (Fig. 2), with 23.4% (1990s) and 27% (2010s) of flows from the pelagic compartment transferred to the demersal habitat through consumption when predation of planktonic groups on their prey was excluded (results not shown). Results also emphasized the important role of small pelagic fishes (functional group or F.G. 31) in linking TL II from the pelagic compartment with higher trophic levels from the pelagic and demersal habitat. Of this group's biomass, 74.6% and 83.5% was transferred to the pelagic higher trophic levels and 25.4% and 16.5% to the demersal higher trophic levels in 1990s and 2010s, respectively (results not shown).

Most of the energy flows were between TL I, II and III (Fig. 3a and b). Exports, primarily represented by catches, were mainly focused on TL III and IV. An important link between detritus and TL II was found, as flows from the detritus and from the primary producers to TL II were similar in

Table 2
Output estimates of the Israeli Mediterranean continental shelf model for 1990–1994 (1990s) and 2008–2012 (2010s) time periods. TL = trophic level; F = fishing mortality ($year^{-1}$); M2 = predation mortality ($year^{-1}$); M0 = other natural mortality ($year^{-1}$); F/Z = exploitation rate (fishing mortality (F)/total mortality (Z)); FD = flow to detritus ($t \cdot km^{-2} \cdot year^{-1}$).

Functional groups		TL		F		M2		MO		F/Z		FD	
		1990s	2010s	1990s	2010s	1990s	2010s	1990s	2010s	1990s	2010s	1990s	2010s
1	Phytoplankton	1.00	1.00	–	–	21.21	18.10	67.21	60.79	0.00	0.00	186.77	184.78
2	Benthic primary producers	1.00	1.00	–	–	2.75	3.85	2.75	1.65	0.00	0.00	0.22	0.11
3	Micro and Mesozooplankton	2.05	2.05	–	–	21.68	22.03	1.14	1.16	0.00	0.00	33.06	30.72
4	Macrozooplankton	2.77	2.77	–	–	16.29	16.56	0.86	0.87	0.00	0.00	3.74	3.66
5	Gelatinous plankton	2.89	2.89	–	–	4.67	4.75	10.89	11.07	0.00	0.00	1.53	1.46
6	Polychaetes	2.05	2.05	–	–	4.23	4.30	1.06	1.07	0.00	0.00	7.10	7.41
7	Suprabenthos	2.16	2.16	–	–	9.47	9.50	2.37	2.38	0.00	0.00	4.13	4.41
8	Native shrimps	2.97	2.97	0.43	0.44	2.52	2.51	0.16	0.16	0.14	0.14	0.25	0.17
9	Alien shrimps	2.96	2.96	0.39	0.41	2.55	2.55	0.16	0.16	0.13	0.13	0.14	0.23
10	Native crabs	2.89	2.89	0.01	0.01	2.65	2.66	0.14	0.14	0.00	0.00	0.09	0.04
11	Alien crabs	2.89	2.90	0.56	0.72	2.10	1.95	0.14	0.14	0.20	0.26	0.05	0.19
12	Benthic invertebrates	2.15	2.16	–	–	2.62	2.62	0.65	0.66	0.00	0.00	3.20	3.44
13	Benthic cephalopods	3.29	3.33	0.12	0.15	1.69	1.66	0.09	0.10	0.06	0.08	0.04	0.05
14	Benthopelagic cephalopods	3.57	3.58	0.23	0.31	2.14	2.07	0.13	0.13	0.09	0.12	0.35	0.36
15	Mullets	3.10	3.11	1.02	0.89	0.76	0.95	0.08	0.03	0.55	0.48	0.09	0.01
16	Goatfishes	3.11	3.14	1.01	0.98	0.86	0.88	0.02	0.02	0.54	0.52	0.05	0.05
17	Hake	4.05	4.06	0.47	0.26	0.42	0.61	0.00	0.02	0.53	0.29	0.05	0.01
18	Flatfishes	3.28	3.29	0.69	0.65	0.79	0.81	0.04	0.06	0.45	0.43	0.03	0.04
19	Rocky fishes	3.02	3.03	0.26	0.41	1.38	1.28	0.08	0.04	0.15	0.23	0.09	0.08
20	Small native dem. fishes	3.25	3.29	0.66	0.78	0.80	0.65	0.03	0.07	0.44	0.52	0.32	0.30
21	Large native dem. fishes	3.70	3.80	0.80	0.78	0.27	0.29	0.04	0.04	0.72	0.71	0.13	0.03
22	Alien herbivores	2.00	2.00	0.24	0.36	1.40	1.28	0.09	0.09	0.14	0.21	0.01	0.03
23	Earlier alien dem. fishes	3.25	3.26	0.58	0.61	0.99	0.93	0.01	0.05	0.37	0.38	0.04	0.15
24	New alien dem. fishes	–	3.35	–	0.64	–	0.77	–	0.08	–	0.43	–	0.15
25	Alien lizardfish	3.78	3.87	0.78	0.78	0.32	0.31	0.09	0.08	0.66	0.66	0.10	0.05
26	Demersal fishes (upper slope)	3.70	3.70	0.11	0.09	0.86	0.84	0.04	0.08	0.11	0.09	0.01	0.02
27	Benthopelagic fishes	3.14	3.14	0.31	0.43	1.62	1.52	0.10	0.10	0.15	0.21	0.32	0.28
28	Mesopelagic fishes	3.27	3.27	0.00	–	1.65	1.66	0.09	0.09	0.00	0.00	0.11	0.08
29	Demersal sharks	4.04	4.10	0.36	0.46	0.08	0.03	0.25	0.21	0.52	0.66	0.02	0.04
30	Rays and skates	3.59	3.62	0.70	0.74	0.12	0.09	0.12	0.11	0.75	0.79	0.03	0.06
31	Small pelagic fishes	3.07	3.07	0.18	0.08	2.09	2.25	0.12	0.12	0.08	0.03	1.76	1.49
32	Horse mackerel	3.30	3.32	0.48	0.46	0.89	0.93	0.07	0.07	0.33	0.31	0.22	0.15
33	Mackerel	3.53	3.53	0.26	0.31	1.09	1.06	0.07	0.07	0.18	0.22	0.12	0.13
34	Native medium pelagic fishes	4.13	4.15	0.41	0.58	0.35	0.20	0.09	0.09	0.49	0.67	0.15	0.11
35	Alien medium pelagic fishes	–	4.12	–	0.46	–	0.31	–	0.09	–	0.54	–	0.04
36	Large pelagic fishes	4.25	4.26	0.83	0.86	0.05	0.04	0.01	0.00	0.92	0.95	0.07	0.07
37	Turtles	3.05	3.05	0.08	0.09	0.00	0.00	0.08	0.09	0.50	0.50	0.04	0.04
38	Sea birds	3.04	3.05	0.04	0.04	0.00	0.00	5.06	5.06	0.01	0.01	0.02	0.02
39	Dolphins	4.33	4.30	0.04	0.04	0.00	0.00	0.05	0.05	0.48	0.48	0.02	0.02
40	Detritus	1.00	1.00	–	–	–	–	–	–	–	–	0.00	0.00
41	Discards	1.00	1.00	–	–	–	–	–	–	–	–	0.00	0.15

both time periods. The Transfer Efficiencies (TE), decreased with the increasing TL (Fig. 3). The mean TE was 18.9% and 19% for the two time periods, respectively (Table 3).

Analyses of flows in terms of the Lindeman spine for both time periods indicated similar patterns when studying the overall food web (Fig. 3a and b). However, results from different habitats (pelagic versus demersal), highlight a change in the food web over time (Fig. 4): biomass, production and consumption of pelagic groups located in TL III decreased between the two time periods while values for TL IV remained constant. In contrast, the biomass, production and consumption of demersal groups located in TL III increased between the two time periods while values at TL IV decreased (Fig. 4).

3.2. Ecological roles of functional groups

In the 1990s, functional groups ranged from TL = 1 for primary producers (F.G. 1 and 2) and detritus (F.G. 40–41) to TL = 4.33 for dolphins (F.G. 39) (Table 2). The highest TLs (>4) corresponded to dolphins (F.G. 39), hake (F.G. 17), demersal sharks (F.G. 29), native and alien medium pelagic fishes (F.G. 34 and 35) and large pelagic fishes (F.G. 36). Invertebrates were classified with a TL between 2.05 and 2.97, with cephalopods (F.G. 13 and 14) showing higher TLs. Fish had TLs between 3.02 and 4.25, with the exception of the alien herbivores (F.G. 22), which had a lower TL (TL = 2). Similar TLs were found for the 2010s model. TLs estimated from the 2010s model were highly and positively correlated with the ^{15}N values from Lebanon (Fig. 5, Spearman-rank correlation coefficient, $R_s = 0.90$, $n = 12$, $p < 0.0001$). These results showed that an increase of ^{15}N values calculated from SIA coincided with an increase of TLs estimated by the Ecopath model.

The MTI analysis showed that most of the groups had a negative impact on themselves as a result of competition for the resources within the group and had a direct negative impact on their main prey through predation (Fig. 6a and b). Results highlighted some indirect effects between groups through competition for resources. For example, between benthic invertebrates (F.G. 12) and alien herbivores (F.G. 22), and between small native demersal fishes (F.G. 20), earlier alien demersal fishes (F.G. 23) and new alien demersal fishes (F.G. 24). An important impact of small pelagic fishes (F.G. 31) and small native demersal fishes (F.G. 20) on higher and lower trophic levels was also observed. A comparison between time periods highlighted the increasing impact of demersal sharks (F.G. 29), rays and skates (F.G. 30) and alien groups: earlier alien dem. fishes (F.G. 23), new alien dem. fishes (F.G. 24) and alien medium pelagic fishes (F.G. 35). By contrast there was a decreasing impact of mullets (F.G. 15), hake (F.G. 17), large native demersal fishes (F.G. 21), alien lizardfish (F.G. 25) and native medium pelagic fishes (F.G. 34) due to changes in biomass composition.

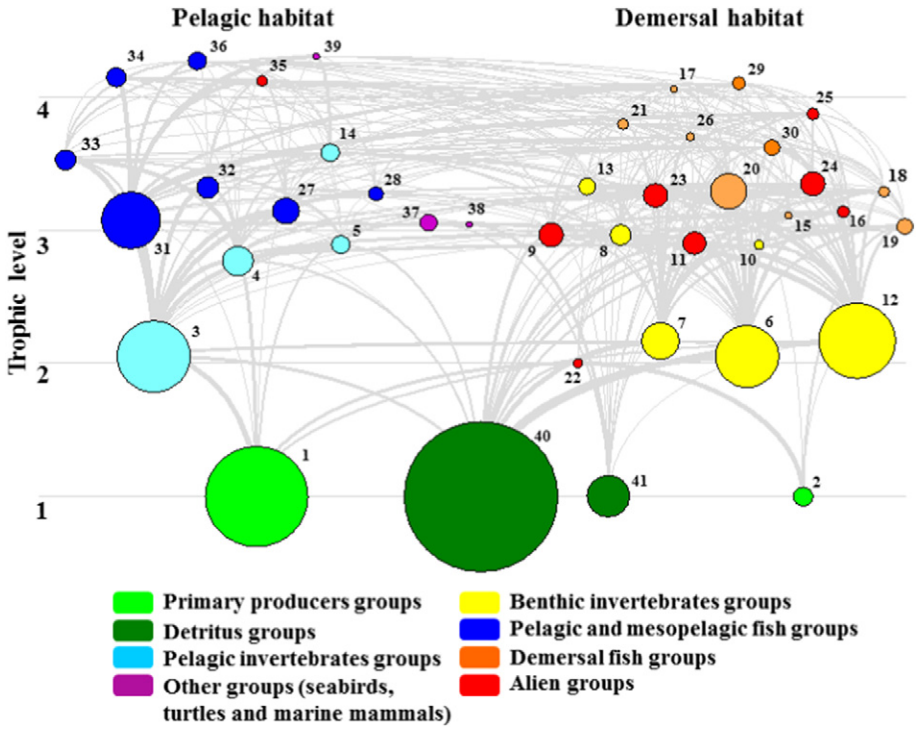


Fig. 2. Flow diagram of the Israeli Mediterranean continental shelf model representing the 2008–2012 period. The size of each circle is proportional to the biomass of the functional group. The numbers identify the functional groups of the model (listed in Table 1). The thickness of the connecting lines is proportional to the magnitude of their trophic flows.

Native medium pelagic fishes (F.G. 34), dolphins (F.G. 39), demersal sharks (F.G. 29), large pelagic fishes (F.G 36), alien lizardfish (F.G. 25) and squids (F.G. 14) may have been potential keystone species in the ICS in both time periods (Fig. 7a and b). Hake (F.G. 17) was identified as a potential keystone species during the 1990s but not in the 2010s. This result indicates that the keystone role of hake in the ICS ecosystem disappeared in the latter period.

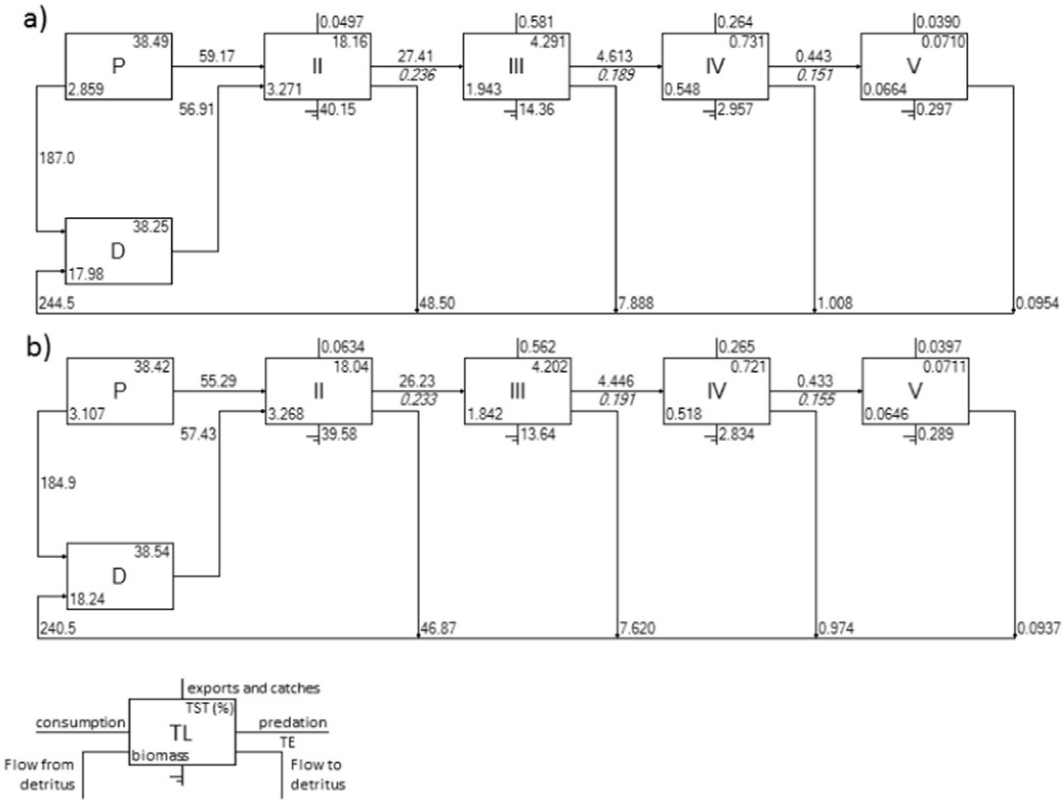


Fig. 3. Lindeman spine representation of the Israeli Mediterranean continental shelf ecosystem for the 1990–1994 (a) and the 2008–2012 (b) time periods. Trophic level (TL) I is split into primary producers (P) and detritus (D). Flows are represented in $t \cdot km^{-2} \cdot year^{-1}$ and biomass in $t \cdot km^{-2}$.

Table 3
Characteristics, statistics and ecological indicators for the Israeli Mediterranean continental shelf model (ICS) in the two time periods.

Indicators	1990–1994	2008–2012	Units
Number of functional groups	39	41	
Number of primary producers	2	2	
Number of alive functional groups	37	39	
Total system throughput (TST)	646.36	631.89	$t \cdot km^{-2} \cdot year^{-1}$
Total primary production/Total respiration (Pp/R)	4.26	4.26	
Total primary production/Total biomass (Pp/B)	28.32	27.28	
System omnivory index (SOI)	0.19	0.19	
Finn's cycling index (of total throughput) (FCI)	5.72	5.78	% of TST
Finn's mean path length (PL)	2.62	2.63	
Total biomass (excluding detritus) (TB)	8.69	8.80	$t \cdot km^{-2}$
Mean trophic level of the community (mTLco)	1.35	1.34	
mTLco (excluding TL = 1)	2.62	2.60	
Total catches (TC)	0.94	0.93	$t \cdot km^{-2} \cdot year^{-1}$
Total landings	0.80	0.64	$t \cdot km^{-2} \cdot year^{-1}$
Total discards	0.14	0.29	$t \cdot km^{-2} \cdot year^{-1}$
Mean trophic level of the catch (mTLc)	3.38	3.37	
Primary production required to sustain the fisheries (PPR, considering PP + detritus)	10.36	11.34	%
Mean transfer efficiency (TE)	18.90	19.00	%
Ecopath pedigree index	0.54	0.54	

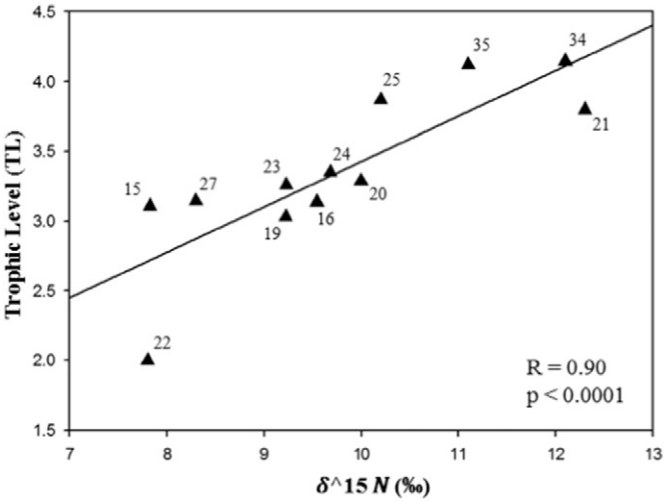


Fig. 5. Correlation between trophic level (TL) calculated with the 2008–2012 Ecopath model (2008–2012 period) and the ^{15}N values calculated from stable isotope analysis conducted along the Lebanese coast (Fanelli et al., 2015). The numbers in the figure identify the functional groups of the model (listed in Table 1).

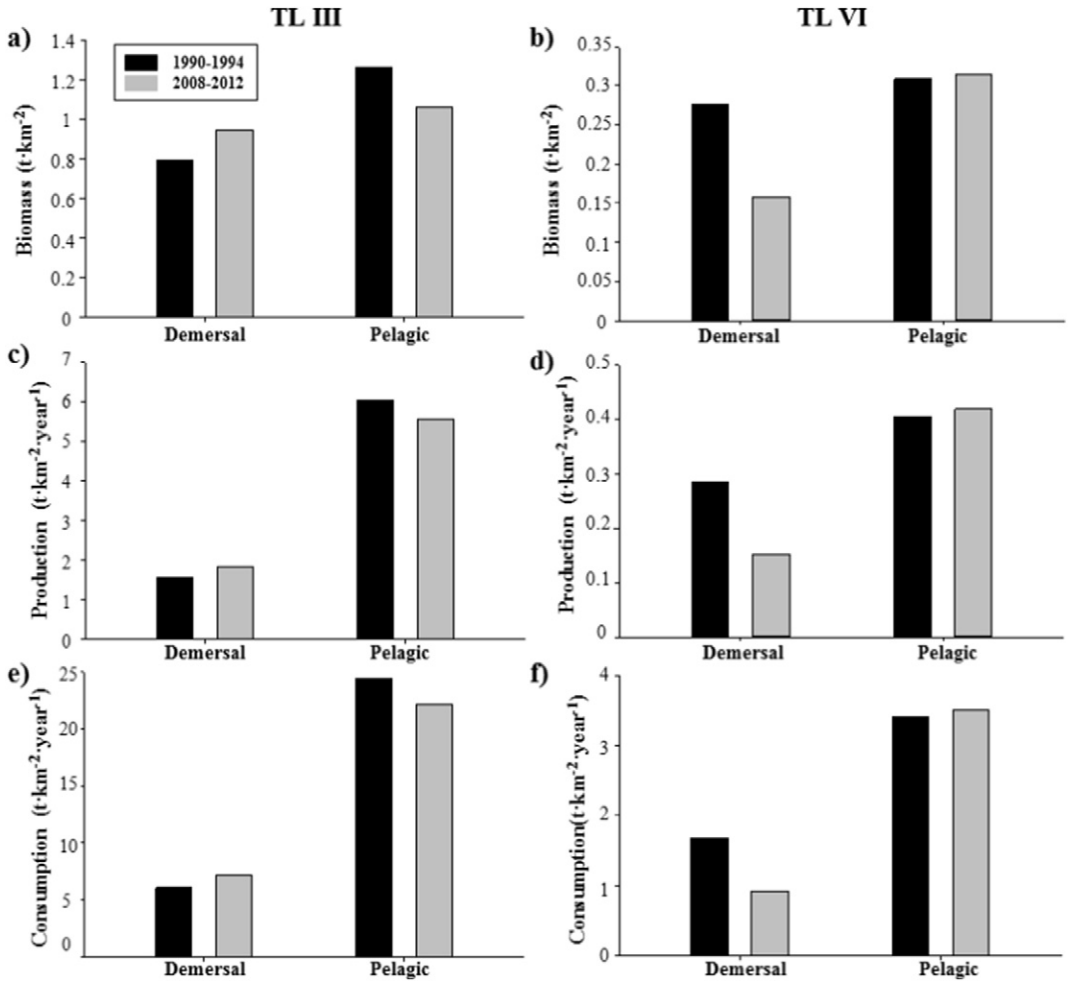


Fig. 4. Biomass ($t \cdot km^{-2}$), production ($t \cdot km^{-2} \cdot year^{-1}$) and consumption ($t \cdot km^{-2} \cdot year^{-1}$) by discrete trophic levels (TL) and habitat in both time periods. Left hand panels are TL III and right panels are TL IV. Top row is biomass (a and b), middle row is production (c and d) and bottom row is consumption (e and f).

3.3. Impacts and role of alien species

The contribution of alien groups to the total biomass increased from 9.5% in 1990s to 23.3% in 2010s (Fig. 8a). For crustaceans (shrimps and crabs) and demersal fishes the alien species increased from 35.1% and 20.2% to 66.3% and 47.5%, respectively (Fig. 8a). For example, alien shrimps (F.G. 9) and earlier alien demersal fishes (F.G. 23) increased their biomass by 55.5% and 256.6%, respectively (Table 1). A remarkable result is the large biomass of new alien demersal fishes (F.G. 24) achieved within a decade of the invasion of the first species of this group, with a rate of increase of 0.01 $t \cdot km^{-2} \cdot year^{-1}$. The biomass of alien pelagic fish species increased from 0% to 2.6% between the 1990s and 2010s (Fig. 8a).

The analysis of the catch showed similar patterns to those observed in the biomass. The contribution of alien groups to the total catch increased from 14.9% in 1990s to 34.1% in 2010s (Fig. 8b). This increase was from 43.1%, 22.2% and 0% to 75.4%, 46.1% and 4.9% for crustaceans, demersal fishes and pelagic fishes, respectively (Fig. 8b). The catch of alien crabs (F.G. 11) and alien herbivores (F.G. 22) increased 399.1% and 224.3%, respectively (Table 1). The analysis of the percent contribution of alien species to the catch for each fleet showed that the largest increases corresponded to the trawl and the recreational fleet, with increases from 30.1% and 6.2% to 48% and 18.2% from 1990s to 2010s, respectively (Fig. 8c).

The analysis of predation and consumption (Fig. 9a and b), as well as the MTI analysis, showed the increasing importance of alien groups between the two time periods. Results highlight that the predation by alien species on their prey increased slightly more than their consumption by their predators (Fig. 9a and b). For example, the grazing of alien herbivores (F.G. 22) on primary producers increased 109.8% while the predation on alien herbivores by their predators increased 90.6%.

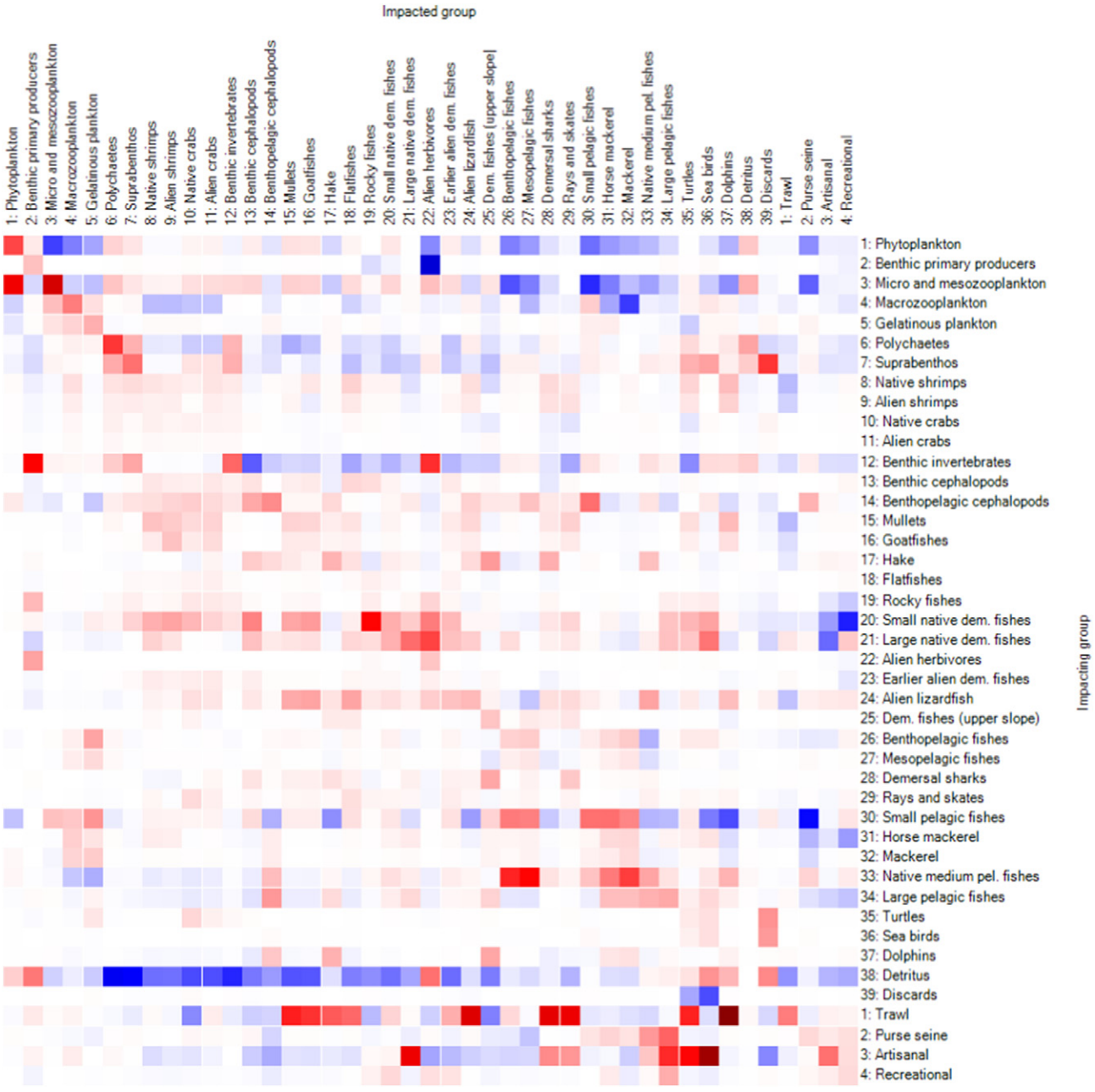


Fig. 6. Mixed Trophic Impact (MTI) analysis of the Israeli Mediterranean continental shelf ecosystem for 1990–1994 (a) and 2008–2012 (b) time periods. Negative (red) and positive (blue) impacts are represented.

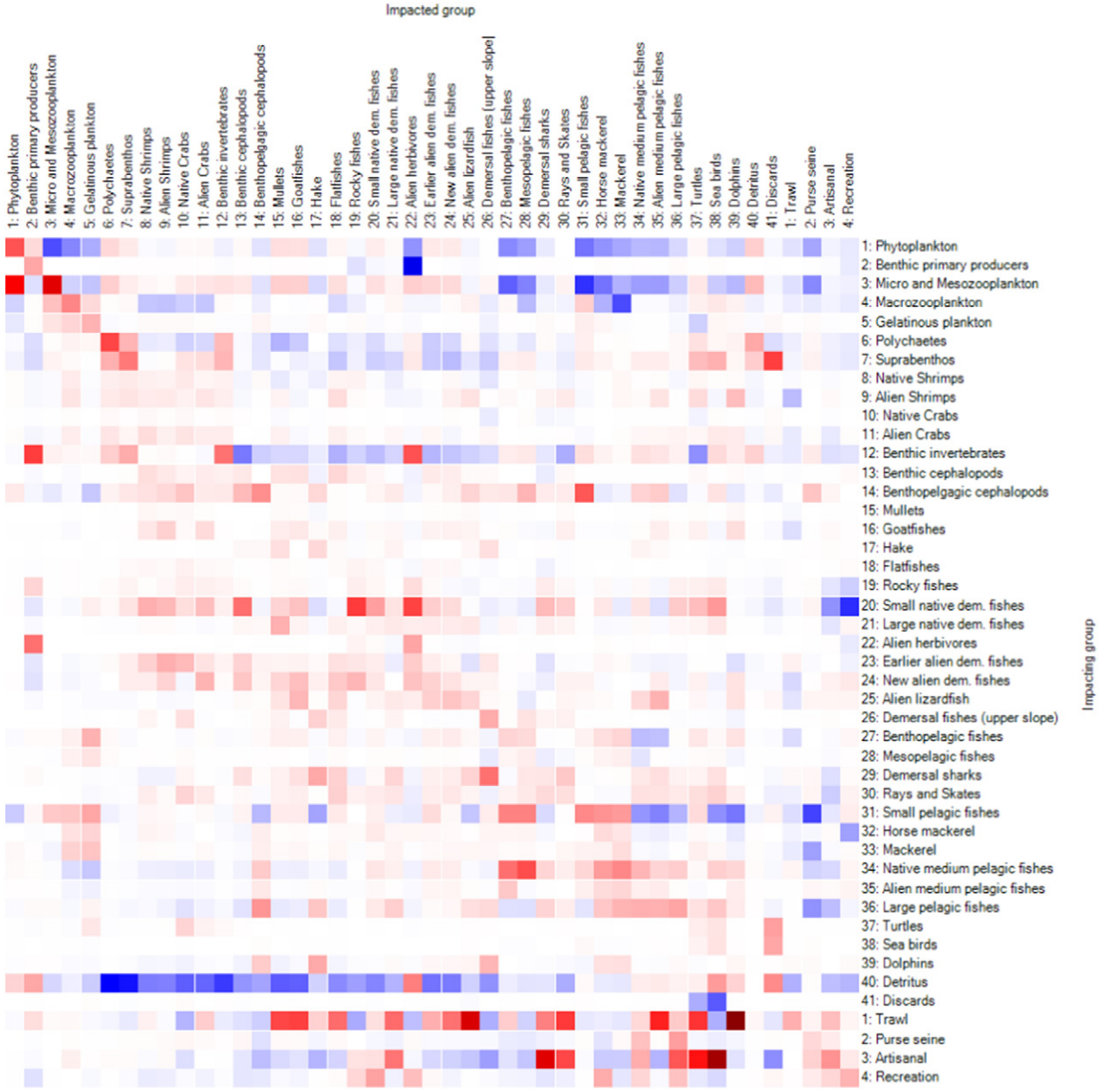


Fig. 6 (*continued*).

The keystoneness index analysis showed that alien lizardfish (F.G. 25) was identified as a potential keystone species in both time periods (Fig. 7a and b). It also showed the possibility of alien medium pelagic fish (F.G. 35) to become a keystone group in the future, as this group had a high keystoneness index despite its recent invasion (Fig. 7b).

3.4. Impacts of fishing on the food web

Total catch in 1990s was 0.94 $t \cdot km^{-2} \cdot year^{-1}$, and did not differ considerably between both time periods, suggesting stable catch production (Table 3). Exploitation rates (F/Z) (Table 2) were high for many of the exploited demersal groups, including, mullets (F.G. 15), goatfishes (F.G. 16), large native demersal fishes (F.G. 21), alien lizardfish (F.G. 25), demersal sharks (F.G. 29), rays and skates (F.G. 30); and medium (F.G. 34 and 35) and large pelagic fishes groups (F.G. 36) (Table 2).

The percentage of primary production required to sustain the fisheries (%PPR) was 10.36% and 11.34% for the 1990s and 2010 time periods, respectively (Table 3).

The MTI analysis applied to the fishing fleets showed that the four fleets included in the ICS models had negative impacts on themselves and to a lesser extent on the other fleets, indicating direct and indirect competition for marine resources (Fig. 6a and b). Most fleets had expected impacts on target species. The trawl fleet had the highest impact on many demersal and pelagic groups and also had a high impact on turtles (F.G. 37) via direct mortality, and on dolphins (F.G. 39) via direct mortality and competition for resources. The artisanal fleet had a high impact on turtles (F.G. 37) and seabirds (F.G. 38) via direct mortality.

4. Discussion

These are the first published food web models of the Israeli Mediterranean continental shelf (ICS) ecosystem. After parameterization, data quality analysis and balancing the model, results showed the important role of alien fish, shrimps and crabs on the southeastern Mediterranean Sea and their increased impact from the 1990s to 2010s. Fishing also had notable impacts on the ecosystem, already in the 1990s as well as in the 2010s.

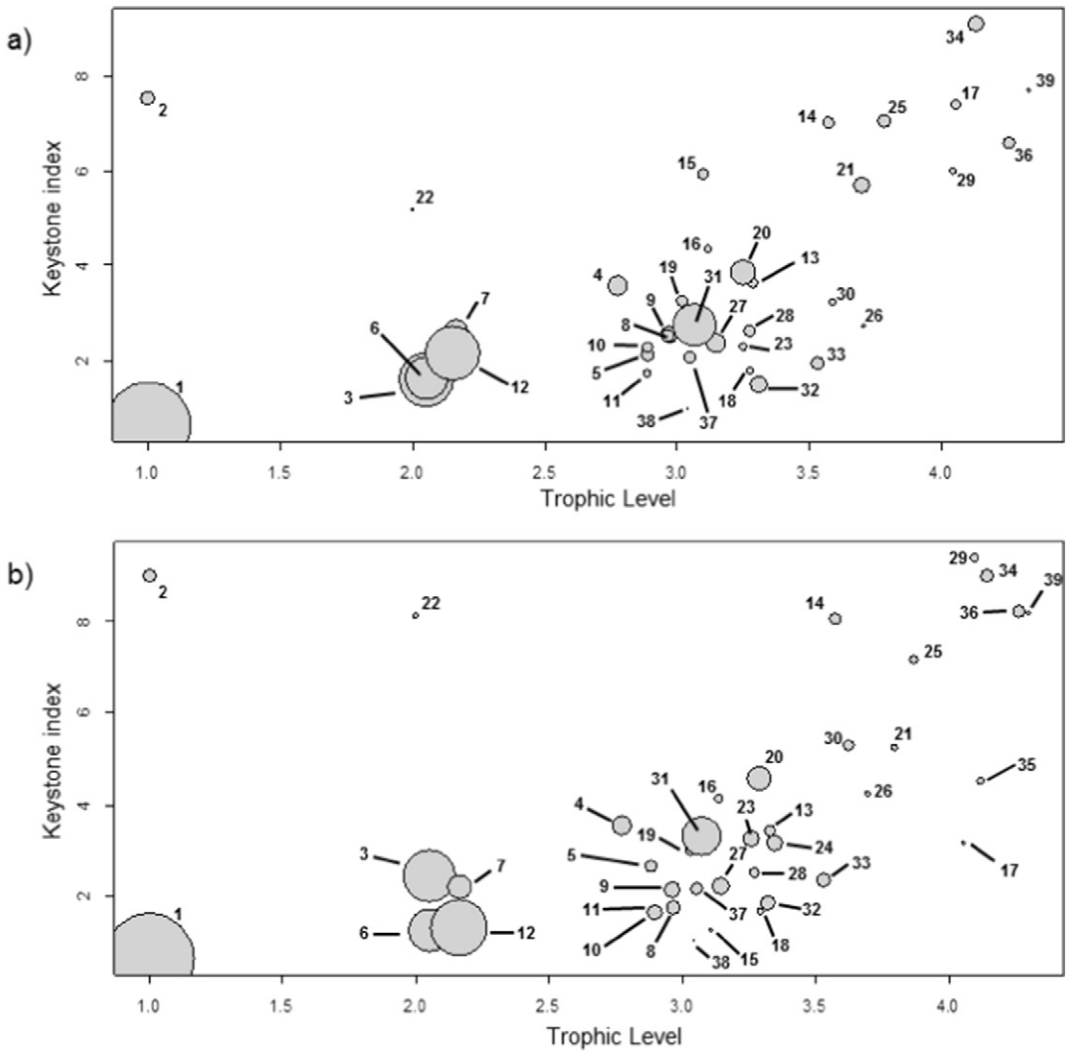


Fig. 7. Functional groups plotted against keystone index and trophic level for 1990–1994 (a) and 2008–2012 (b) time periods. The numbers identify the functional groups of the model (listed in Table 1). The size of each circle is proportional to the biomass of the functional group.

This study represents an important step forward in evaluating the past and current impacts of alien species and fishing activities in the ICS. One of the main advantages of undertaking such a study is the identification of information gaps. The pedigree index obtained for both ICS models (1990s and 2010s) indicated an acceptable quality of the models (Morissette, 2007; Lassalle et al., 2014) although they are among the lowest values in the Mediterranean Sea (Corrales et al., 2015). There are many information gaps, specifically in the pelagic habitat and benthic invertebrates groups. Population assessments of small and medium pelagic fishes and plankton sampling in the study area would greatly improve the parameter estimates of these groups. A fisheries independent annual trawl survey would improve the available data and increase the understanding and quantification of changes in the Levantine Sea. Detailed stomach content analyses are needed for alien species to complement available information and our understanding of the impact of alien species in this area.

Obtaining good estimates of total catch (both official and Illegal, Unregulated and Unreported (IUU)) (Edelist et al., 2013b) was challenging but the reconstruction used in this study offered a more complete estimation than the official data available. However, as Edelist et al. (2013b) indicated, the catch data of the artisanal fleet and the purse seine were less reliable than the trawl catch due to the large number of vessels and landing sites. In addition, the recreational sector requires detailed examination due to its actual importance and its potential growth, as in other Mediterranean regions (Gaudin and De Young, 2007; Pauly et al., 2014).

Comparison and calibration of model outputs are part of the model validation process. Here we used information from Stable Isotope Analysis (SIA) from a neighboring area to compare outputs with trophic levels estimates from the model. Our results showed a high correlation with the SIA estimates obtained for the Lebanese coast (Fanelli et al., 2015), highlighting that the diet information used here represent the trophic relationships reasonably well. Although the stomach content analysis and SIA were not performed in the same area, they are from the Levantine Sea and could be used here to parameterize and validate our models. Future work should include more independent analysis to validate the models (i.e. information on fishing mortalities from stock assessment) while the 1990s model should be calibrated to available time series of historical data using the temporal modeling approach Ecosim (Walters et al., 1997).

4.1. Ecosystem structural and functioning traits and ecological role of functional groups

In agreement with the oligotrophic nature of the Levantine Sea, TST and TB estimates, were much lower in our models than in other Mediterranean models set up in the western and central basin (Corrales et al., 2015). These features had been also observed on a larger scale

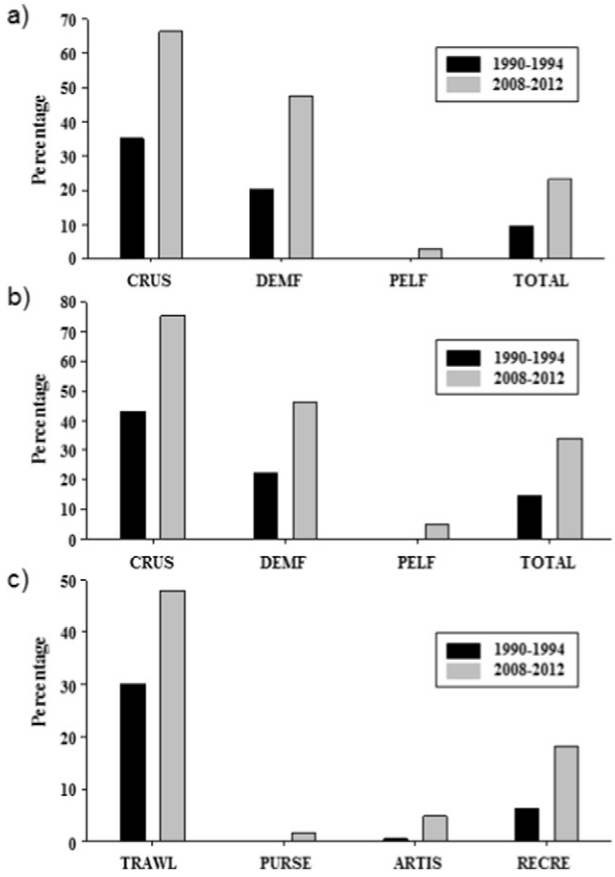


Fig. 8. Contribution (%) of alien species to the total biomass (a), to the catch of different groups (b) and to the total catch of the various fleets (c). Total biomass includes all the groups with sufficient information to split between native and alien species (fish, cephalopods and crustaceans (shrimps and crabs)). Planktonic groups, suprabenthos, polychaetes and benthic invertebrates are excluded for the analysis. CRUS = crustaceans (shrimps and crabs); DEMF = demersal fishes; PELF = pelagic fishes; PURSE = purse seine; ARTIS = artisanal; RECRE = recreational.

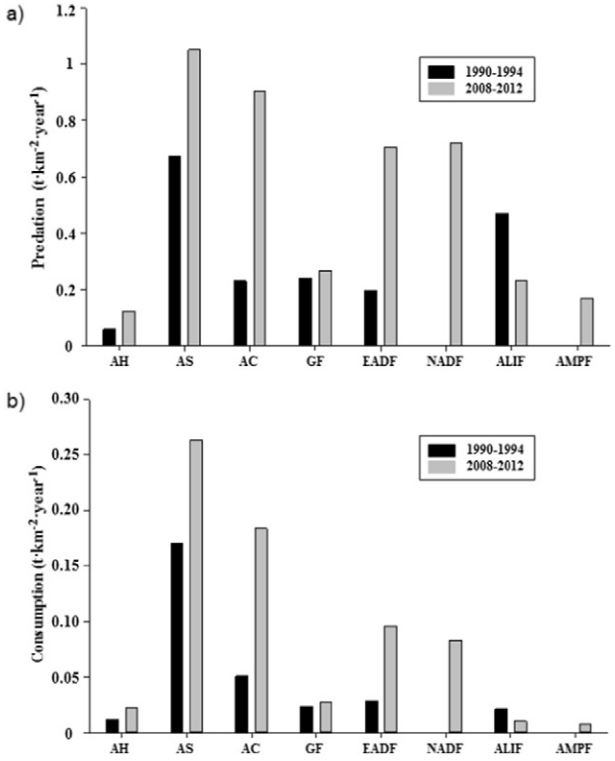


Fig. 9. Predation ($t \cdot km^{-2} \cdot year^{-1}$) by alien species on their prey (a) and consumption ($t \cdot km^{-2} \cdot year^{-1}$) of alien species by their predators (b) for the two time periods (1990–1994 and 2008–2012). AH = alien herbivores; AS = alien shrimps; AC = alien crabs; GF = goatfishes; EADF = earlier alien demersal fishes; NADF = new alien demersal fishes; ALIF = alien lizardfish; AMPF = alien medium pelagic fishes.

Mediterranean Sea food web model (Piroddi et al., 2015). The TEs of both time periods were higher (almost double) than the average value of 10% reported worldwide (Pauly and Christensen, 1995). These values were also higher than in other Mediterranean Sea areas (Corrales et al., 2015) and for the Mediterranean Sea as a whole (Piroddi et al., 2015). Such high values highlight the nutrient-poor waters of the Israeli Mediterranean coast.

Results related with ecosystem development theory (Odum, 1969; Odum, 1971), suggest that the Israeli ecosystem was at a developing stage in both time periods. For example, the Pp/R values were higher than 1, indicating that more energy was produced than respired within the system. The Pp/B values were high, indicating low levels of biomass accumulation within the system compared with productivity; and the FCI values, which represents the proportion of throughput cycled within the ecosystem, was low, indicating an immature system. Food web complexity indices such as the SOI and PL were low, suggesting that the ICS models look more chain-like than web-like. All of these indicators presented similar values in both time periods, suggesting that the functioning of the whole ecosystem did not change significantly over time, and could indicate that the ecosystem was already highly impacted in the 1990s.

Results from the Lindeman spine suggest food web changes in different habitats. The increasing flows in the demersal habitat in TL III can be due to the increasing impact of alien species on the ecosystem. This could be related to the empty niches left by the native species that the alien species occupied or as a result of a potential trophic cascade due to the overexploitation of top predators like hake (F.G. 17), large native demersal fishes (F.G. 21) and alien lizardfish (F.G. 25). However, the decreasing flows in TL III of the pelagic habitat were not clear and should be interpreted carefully since (1) there is a lack of pelagic data and (2) increasing water temperature and river damming were postulated to play an important role in these declines (Edelist et al., 2013b).

The Israeli marine ecosystem was found to share some common features in structure and functioning with other Mediterranean ecosystems such as the important role of detritus via TL II; the dominance of the pelagic fraction in term of flows; and the importance of the benthic-pelagic coupling. For example, the contribution of planktonic groups to the detritus and the relationship between detritus and organisms at TL II (mainly benthic invertebrate groups) was identified as a key process in the Mediterranean Sea (Coll et al., 2006; Tsagarakis et al., 2010; Corrales et al., 2015).

In both time periods the same keystone groups were identified with the exception of hake in 2010s. Most of these groups were also keystones in other Mediterranean areas (Coll and Libralato, 2012; Corrales et al., 2015). Hake (F.G. 17) was not identified as a keystone species in the 2010s model, and could therefore be a native top predator that may have lost its ecological role. The decline of hake has been attributed to overfishing, oceanographic changes, increased temperature and the competition for resources with the alien lizardfish (F.G. 25) (Galil, 2007a; Gucu and Bingel, 2011; Halim and Rizkalla, 2011; Edelist, 2012).

4.2. The role and impact of alien species

Our results show that alien species has become an important part of the ecosystem. Their increasing biomass, catch and flows within the ecosystem, have altered the structure of the food web. Higher percentages of alien species have been found in the catch than in the biomass, as most of the fishing effort has been redirected to the shallow waters

(Edelist et al., 2013b), where most of the commercial demersal biomass is now attributed to alien species (Edelist et al., 2013a).

A high proportion of alien demersal species has been found in other parts of the eastern Mediterranean Sea as well (Gücü and Bingel, 1994; Taşkavak et al., 1998; Harmelin-Vivien et al., 2005; Carpentieri et al., 2009; Gücü et al., 2010). The rapid expansion of the new alien demersal fishes (F.G. 24) is in line with studies that indicated that the most recent wave of invasion has established large populations along the eastern Mediterranean Sea (Edelist et al., 2012; Nader et al., 2012; Edelist et al., 2013a; Stern et al., 2014). The trawl fleet and recreational fishers caught the highest percentages of alien species because most of the trawl fishing effort are focused on shallow soft bottoms; and the recreational fishers catch on hard coastal bottoms, where alien herbivores (F.G. 22) constitute an important part of the total biomass (Goren and Galil, 2001). Except alien herbivores, alien species appear to be poorly established in the rocky littoral zone (Golani et al., 2007). However, the importance of alien species in the catch of all the fleets might be underestimated, especially in the artisanal and the purse seine, as an important part of the catch includes groups that were not sufficiently separate into specific functional groups in the model due to lack of data (e.g. small pelagic fishes and native medium pelagic fishes).

Trophic flows related to alien species increased as a result of their biomass expansion and impacted the ecosystem. The rapid growth of the alien herbivores population altered the community structure of the rocky infralittoral (Sala et al., 2011; Vergés et al., 2014). Prior to their arrival, the role of native herbivorous fishes was negligible, and therefore, the alien herbivores increased the rate of algal recycling and provided more food to potential predators (Galil, 2007b). Moreover, some competition for resources has been found in the demersal habitat. Our results highlighted that the consumption of prey by alien species increased more than the consumption on alien species by their predators. This may be due to the increasing importance of alien species in the catch, preventing energy transfer to higher trophic levels.

Several hypotheses have been proposed to explain the impact that alien species have on the marine ecosystem of the eastern Mediterranean Sea and the decline of native species. Trophic interactions between native and alien species may cause native species to be outcompeted or partially displaced by the invaders (Galil, 2000; Galil, 2007a). Several authors indicated the success of some alien species could be related to the existence of underexploited niches in the ecosystem that could be exploited by the alien species (Golani, 1998; Galil, 2008; Rilov and Galil, 2009). In addition, the impact of fishing activity could facilitate the establishment and spread of alien species as a result of the overexploitation of native species (Galil, 2008) and the possible better adaptation and competitiveness of alien species to proliferate in the highly impacted trawled areas (Edelist et al., 2011). Moreover, environmental conditions (specially temperature but also salinity) might have become more suitable for thermophilic species like most of the Lessepsian migrants, providing various advantages when competing with temperate native species (Golani, 1998; Galil, 2007a).

4.3. Impacts of fishing on the food web

Results of this study showed that fishing activity played an important role over time on the ecosystem and had noticeable impacts. For example, some groups had higher exploitation rates than the general reference point of 0.5 proposed by Rochet and Trenkel (2003), suggesting the overexploitation of several marine resources.

However, the primary production required to sustain the fisheries (%PPR) showed a lower value than the 24.2% estimated for non-tropical shelves worldwide (Pauly and Christensen, 1995). It was lower than the northern and central Adriatic Sea model (Coll et al., 2007), and higher than the south Catalan Sea model (Coll et al., 2006) and the north Aegean Sea model (Tsagarakis et al., 2010). The total catch was stable between both periods, but landings decreased around 20% from 1990s to 2010s while discards increased dramatically, corroborating the unsustainability of the Israeli fisheries (Goren et al., 2013; Edelist et al., 2014). There are reasons for this: The trawl fleet refocused the effort to the shallow waters, where many fish species are characterized by small size and where many nursery habitats are found; and many of the new alien species, found mainly in shallow waters, are discarded as they are small or are venomous/poisonous or for cultural reasons (Edelist et al., 2013b).

The mTLc and mTLco values in the ICS models were higher than models developed in the western and central Mediterranean Sea, and similar to the north Aegean Sea model (Tsagarakis et al., 2010; Coll and Libralato, 2012; Corrales et al., 2015). This could be related to the oligotrophic nature of the eastern Mediterranean Sea and therefore the lower proportion of low TL organisms such as small pelagic fishes in the total catch and the ecosystem, as also seen in the Aegean Sea (Tsagarakis et al., 2010). Moreover, the mTLc and mTLco in the Israeli model are similar during both time periods as the biomass and catch of top predators (i.e., hake (F.G. 17), large native demersal fishes (F.G. 21) alien lizardfish (F.G. 25)) and low trophic levels (like small pelagic fishes (F.G. 31)) decrease while the biomass and catch of medium trophic levels (i.e. earlier (F.G. 23) and new alien demersal fishes (F.G. 24)) increased. In fact, Edelist et al. (2013a) suggested that alien species have masked changes to the mTLco in the demersal fish community by replacing native by alien fishes with similar ecological position in the food web.

4.4. Concluding remarks

This study presents the first attempt to develop a food web model of the Israeli Mediterranean ecosystem. Furthermore it demonstrates the large changes that have occurred to the food web structure and functioning since 1990s. These changes are a consequence of fishing and the increasing number and biomass of alien species.

The current state of the ICS ecosystem is a result of the cumulative impacts of alien species, overfishing and climate change superimposed on the geological history and environmental conditions of the eastern Mediterranean Sea (Galil, 2008; Bianchi et al., 2014). It is difficult to quantify the contribution of each of these factors (Galil, 2007b) but they could be amplified by their cumulative effects. The nature of these effects (additive, synergistic or antagonist) is unknown, although some authors suggested synergistic effects between alien species, fishing and climate change (Galil, 2008; Goren et al., 2013).

Future work including the calibration and fitting of the 1990s model to historical time series data (Walters et al., 1997; Christensen and Walters, 2004) should be used to continue this analysis and evaluate both historical and potential future cumulative impacts of multiple stressors to the ecosystem dynamics of the southeastern Mediterranean Sea.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.jmarsys.2017.02.004.

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